

# Lab: Game Plan -- Mapping Your Future Moves

## Objective

**Learning Goal:** Practice planning as imagining future scenarios, weighing risks and rewards, and building decision trees to navigate uncertainty.

This is a **group challenge** designed for middle school students (Grades 6-8).

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## Materials Needed

- Large paper or whiteboard
  - Colored markers
  - Sticky notes
  - Dice (one per group, for randomness)
  - Lab journal or notebook
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## Part 1: Decision Tree Challenge (15 min)

A decision tree maps out possible futures. Each branch is a choice, and each leaf is an outcome.

**Scenario:** It is Friday afternoon. You have a big project due Monday, a friend invited you to hang out Saturday, and there is a sports tryout Sunday morning.

Build a decision tree on your poster paper. Start with Friday evening and map out all your possible choices and outcomes through the weekend.

Use sticky notes for each decision point and draw arrows for each possible choice.

| Decision Point     | Option A | Option B | What you predict will happen |
|--------------------|----------|----------|------------------------------|
| Friday evening     |          |          |                              |
| Saturday morning   |          |          |                              |
| Saturday afternoon |          |          |                              |
| Sunday morning     |          |          |                              |

*Write your response here*

## Part 2: Risk vs. Reward Analysis (10 min)

Every plan involves trade-offs. Rate each option on a scale of 1-10 for both risk and reward.

| Plan                                                         | Risk<br>(1-10) | Reward<br>(1-10) | Uncertainty<br>(1-10) | Would you<br>choose it? |
|--------------------------------------------------------------|----------------|------------------|-----------------------|-------------------------|
| Do the whole project Friday night, then be free all weekend  |                |                  |                       |                         |
| Split the project across all three days in small chunks      |                |                  |                       |                         |
| Skip the project, have fun, and cram Sunday night            |                |                  |                       |                         |
| Do the project Saturday, skip hanging out, do tryouts Sunday |                |                  |                       |                         |

*Write your response here*

## Part 3: Planning Under Uncertainty (10 min)

Real planning is hard because the future is uncertain. Let's add randomness.

**Dice roll challenge:** For each plan from Part 2, roll a die. If you roll 1-2, something goes wrong (unexpected obstacle). If you roll 3-4, things go as expected. If you roll 5-6, something unexpectedly good happens.

| Plan   | Dice Roll | What happened? | How did this change the outcome? | Could you adapt? |
|--------|-----------|----------------|----------------------------------|------------------|
| Plan 1 |           |                |                                  |                  |
| Plan 2 |           |                |                                  |                  |
| Plan 3 |           |                |                                  |                  |
| Plan 4 |           |                |                                  |                  |

*Write your response here*

## Part 4: Your Real Plan (5 min)

Think about something you actually need to plan for in the next two weeks (a test, a project, a social event, a personal goal).

| Planning Element                                     | Your Plan |
|------------------------------------------------------|-----------|
| What is the goal?                                    |           |
| What are the key decision points?                    |           |
| What is your biggest uncertainty?                    |           |
| What is your Plan B if something goes wrong?         |           |
| What information would help reduce your uncertainty? |           |

*Write your response here*

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## Reflection Table

| Question                                                 | Your Answer |
|----------------------------------------------------------|-------------|
| What is a decision tree?                                 |             |
| Why is planning really about imagining possible futures? |             |
| What makes a plan robust vs. fragile?                    |             |
| How does uncertainty make planning harder?               |             |
| How does Active Inference explain planning?              |             |

*Write your response here*